Indian Vehicle Classification

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Course: IVPAD 2025

Presentation Agenda

Aim of Project

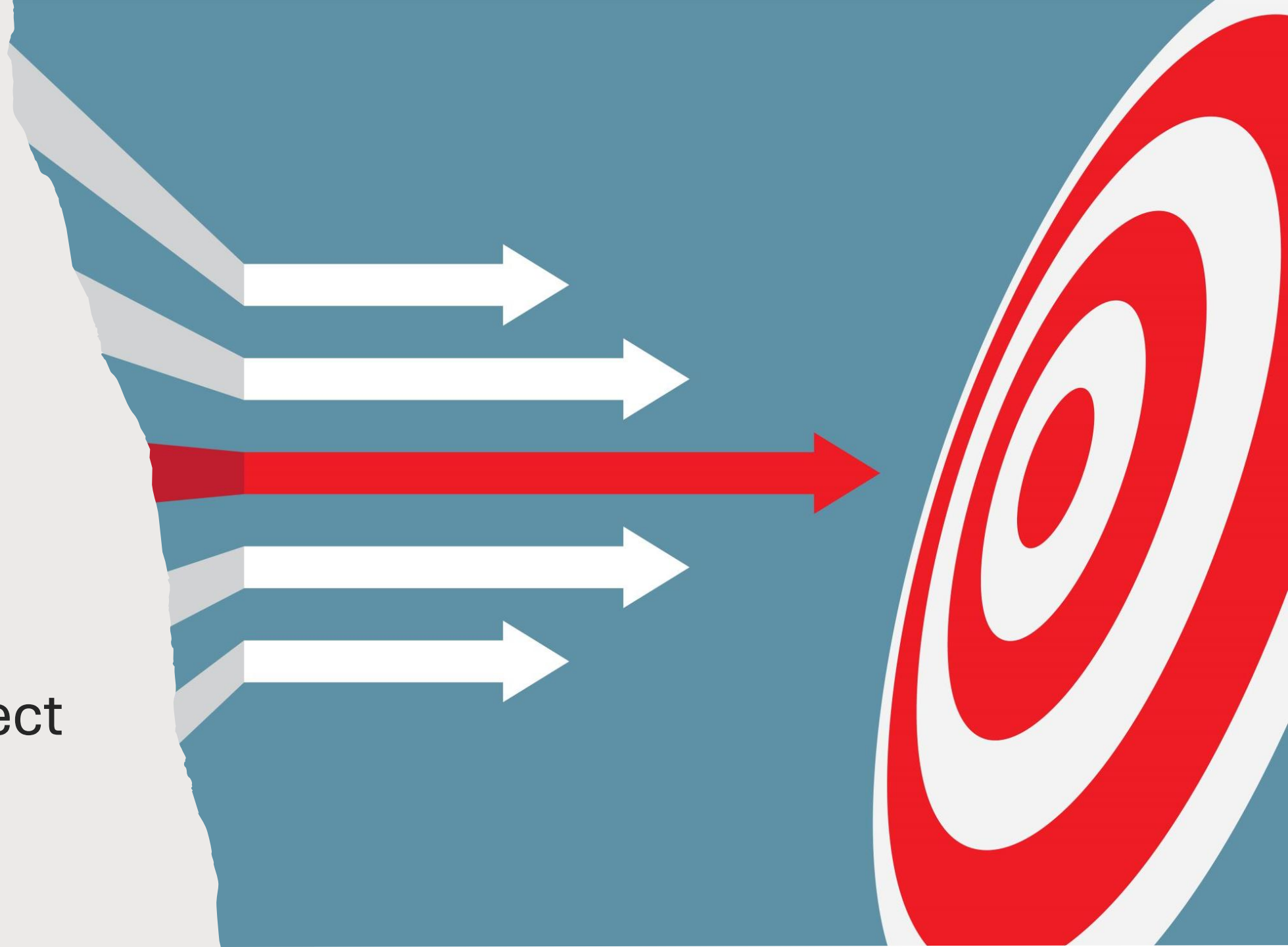
Dataset Overview

Methodology

Performance Analysis

Real-case Study

Aim of project



Aim of the project

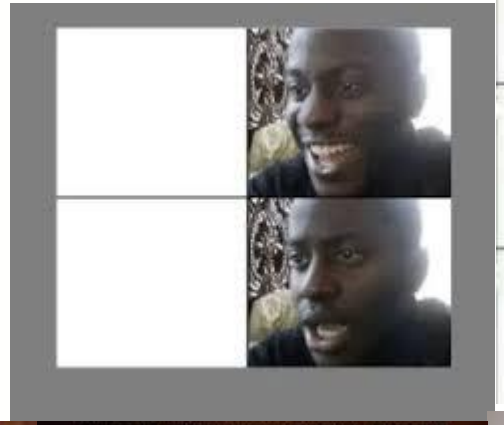
- Classification of the vehicles to 7 classes:

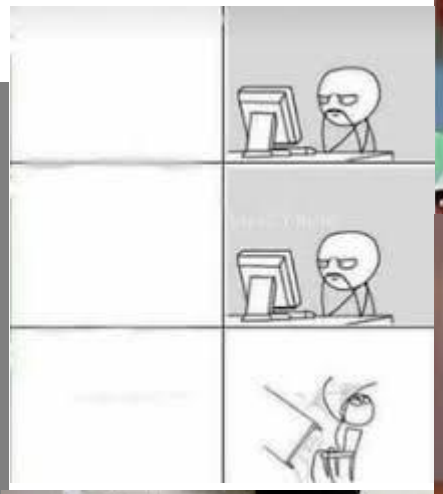
Classes:

Auto
Bike
Car
Motorcycle
Plane
Ship
Train

Scenario

Do you think that Images is important?



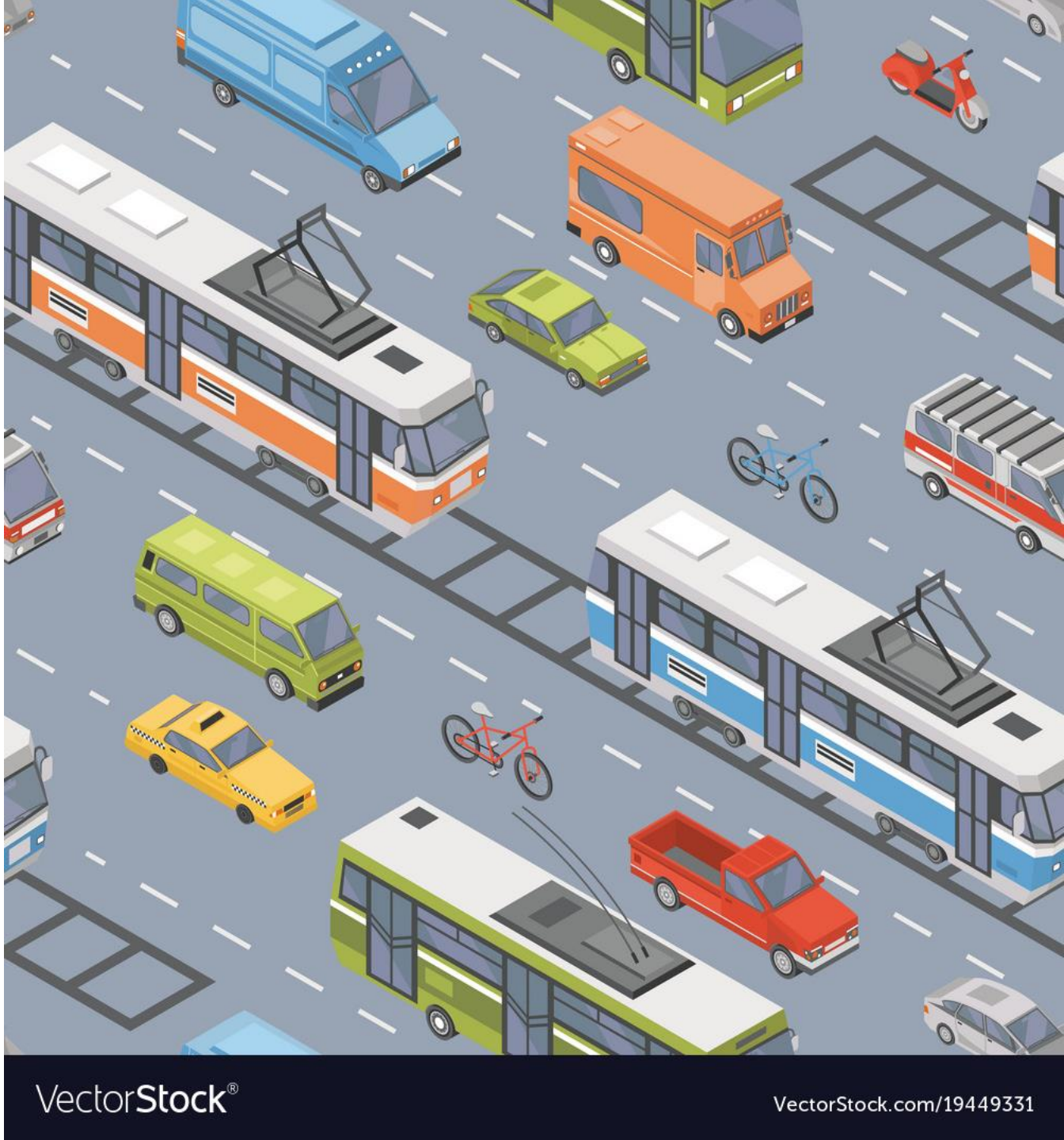


Classification Scenario

Imagine a smart city deploying an **automated traffic monitoring system** using roadside cameras. These cameras are designed to **classify vehicles**—such as cars, motorcycles, buses, trucks, and emergency vehicles—in real time.

Usage:

- Traffic Flow Optimization
- Detecting emergency vehicles
- Lane Enforcement



Motivation

- Automated vehicle classification from images is essential for smart traffic systems, autonomous driving, and urban planning. Traditional methods struggle with accuracy in real-world conditions, while deep learning models can learn complex visual features for reliable classification.
- This project aims to develop an image classification model to accurately identify vehicle types, enhancing traffic monitoring and intelligent transportation solutions.

Dataset Overview

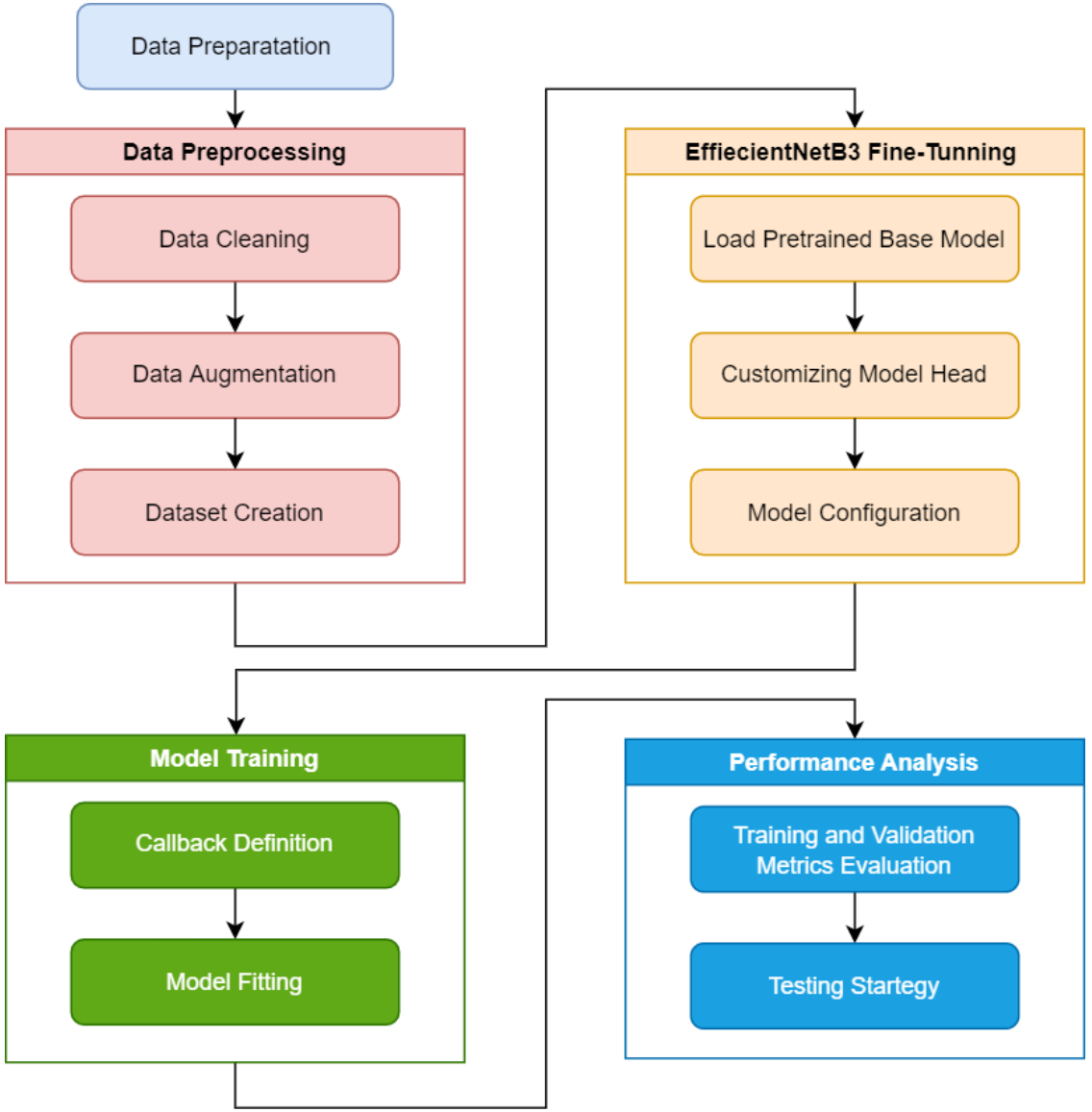
Dataset Overview

- **Overview:** This dataset is designed for vehicle classification tasks and contains a total of 5,600 images distributed across seven categories. Each category represents a different type of vehicle.
- **Image Format:** All images are in JPEG, PNG format with the .jpg and .png extension.
- **folders:**
  - Auto Rickshaws (800 images)
  - Bikes (800 images)
  - Cars (800 images)
  - Motorcycles (800 images)
  - Planes (800 images)
  - Ships (800 images)
  - Trains (800 images)

Source: https://www.kaggle.com/datasets/mohamedmaher5/vehicle-classification

Methodology

Methodology Overview



I. Data Preparation

Objective:
Organize raw image data into a structured format suitable for training, validation, and testing.

Steps Involved:

Data Extraction:
Unzipped the original dataset containing raw image files.

Directory Structuring:
Randomly split the dataset into three subsets:

70% Training
15% Validation
15% Testing

Dataframe Generation:
Created structured DataFrames linking each image to its corresponding label and subset.

II. Data Preprocessing –Data Cleaning

- **Objective:**
  Ensure all images are valid and compatible with the training pipeline by removing corrupted or unsupported files.
- **Cleaning Strategy:**
  - Iterated through all image paths in training, validation, and testing sets.
  - Attempted to open each image and load it fully using PIL.
  - Validated format and mode:
    - Allowed: RGB mode
    - Allowed formats: JPEG, PNG, BMP, GIF
    - Removed: CMYK, WEBP, unreadable, or corrupted files
- **Outcome:**
  - Cleaned dataframes by filtering out flagged entries:
    - train_df_cleaned
    - val_df_cleaned
    - test_df_cleaned

Data Preprocessing – Data Augmentation & Dataset Creation

Data Augmentation

The goal is to enrich training data with made-up variety to improve model generalization.

- **Applied Techniques:**
  - Random Rotation (±20°)
  - Random Zoom (70%–130%)
  - Random Horizontal Flip

Dataset Creation

The goal is to prepare optimized, normalized image datasets for training and validation.

- **Dataset Configuration:**
  - Input dimensions: 320 × 320
  - Batch size: 16 images
  - Normalization: Enabled
- **Labels:**

One-Hot Encoded (categorical mode)

- **Pipeline:**
  - Training set augmented (train_augment)
  - Validation set processed without augmentation

EfficientNetB3 Fine-Tuning

A. Load Base Model

- Pretrained EfficientNetB3 loaded with ImageNet weights
- Top layers excluded → allows custom classification head
- Input shape: (320, 320, 3) for RGB images
- No global pooling applied initially

B. Customizing Model Head

- GlobalAveragePooling2D layer added
- Final Dense layer with 7 units (softmax for multi-class output)
- Architecture wrapped in a Sequential model

C. Model Configuration

- Freeze Pretrained Layers:
  - First 200 layers of base model frozen :retain learned features
  - Remaining layers are trainable: enables task-specific tuning
- Compile Model
  - Loss Function: Categorical Crossentropy
  - Optimizer: Nadam (LR = 1e-4)
  - Evaluation Metric: Accuracy

Model Training

A. Define Callbacks

To improve model performance and prevent overfitting:

- Early Stopping
  - Monitors validation loss
  - Stops training if no improvement for 10 epochs
  - Restores best weights
- Reduce LR On Plateau
  - Reduces learning rate when validation loss plateaus
  - Decrease factor: 0.5
  - Minimum LR: 1e-6
- Model Check point
  - Saves the model with best validation accuracy

B. Train and Validate Model

- Model trained for up to 50 epochs
- Inputting Cleaned & augmented training dataset, Validation dataset
- All callbacks applied to optimize learning

Performance Analysis

Hyperparameter Optimization

Train	Val	test	freezed layer	batch	learning rate
61%	75%	72.69%	350	16	1*10^-4
84.31%	91.37%	87.40%	250	16	1.10^-4
95.09%	93.65%	90.68%	200	16	1*10^-4
93.20%	93.60%	89.50%	200	8	1*10^-4

Training and Validation Metrics Evaluation

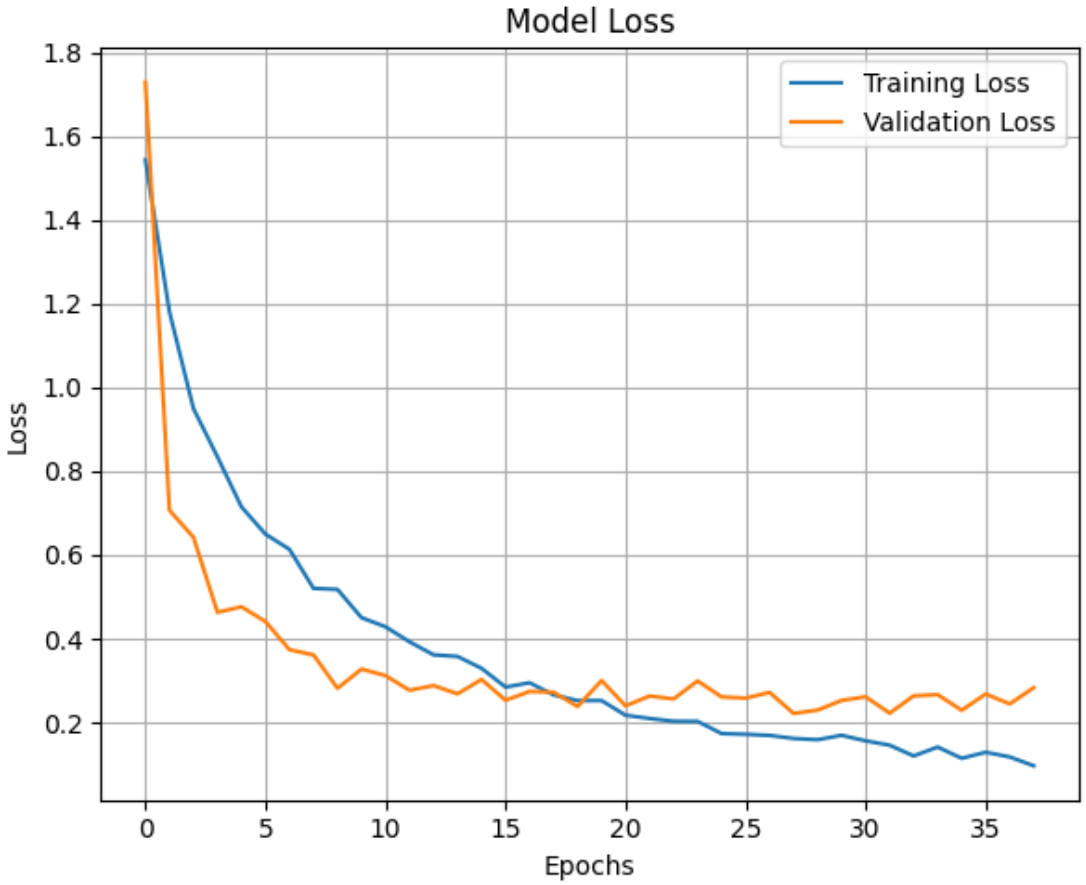
Last epoch

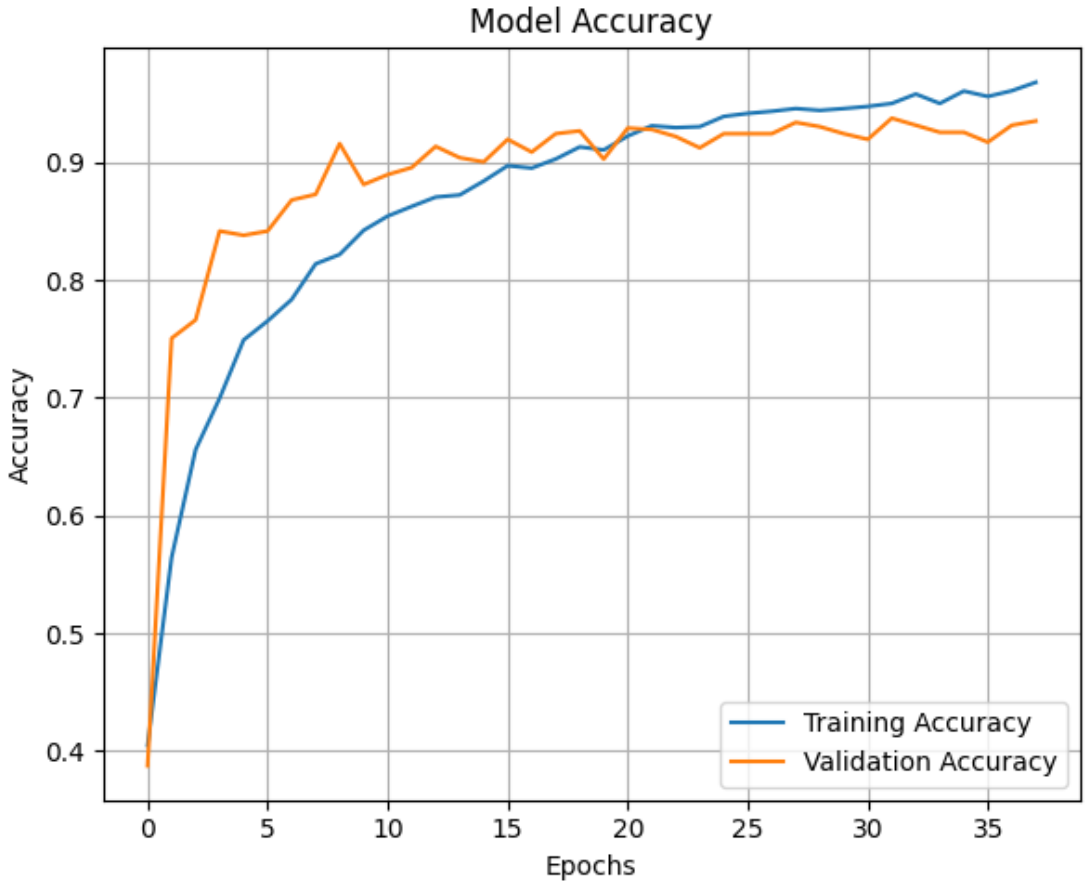
```
Epoch 38: val_accuracy did not improve from 0.93765
244/244 ━━━━━━━━━━━━━━━━━━━━ 111s 453ms/step - accuracy: 0.9688 - loss: 0.0942 - val_accuracy: 0.9353 - val_loss: 0.2830 - learning_rate: 1.0000e-04
Epoch 38: early stopping
Restoring model weights from the end of the best epoch: 28.
<keras.src.callbacks.history.History at 0x79f9fefb7bd0>
```

Best epoch

```
Epoch 28: val_accuracy improved from 0.92926 to 0.93405, saving model to best_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy.
244/244 ━━━━━━━━━━━━━━━━━━━━ 111s 450ms/step - accuracy: 0.9472 - loss: 0.1628 - val_accuracy: 0.9341 - val_loss: 0.2216 - learning_rate: 1.0000e-04
Epoch 29/50
```

Training and Validation Metrics Evaluation





Testing strategy

```
52/52 ━━━━━━━━━━━━━━━━━━━━ 14s 263ms/step - accuracy: 0.9183 - loss: 0.2652
Test loss: 0.30683064460754395
Test accuracy: 0.9105199575424194
```

Internally from dataset:

We have used an image from each class in the test dataset that include:

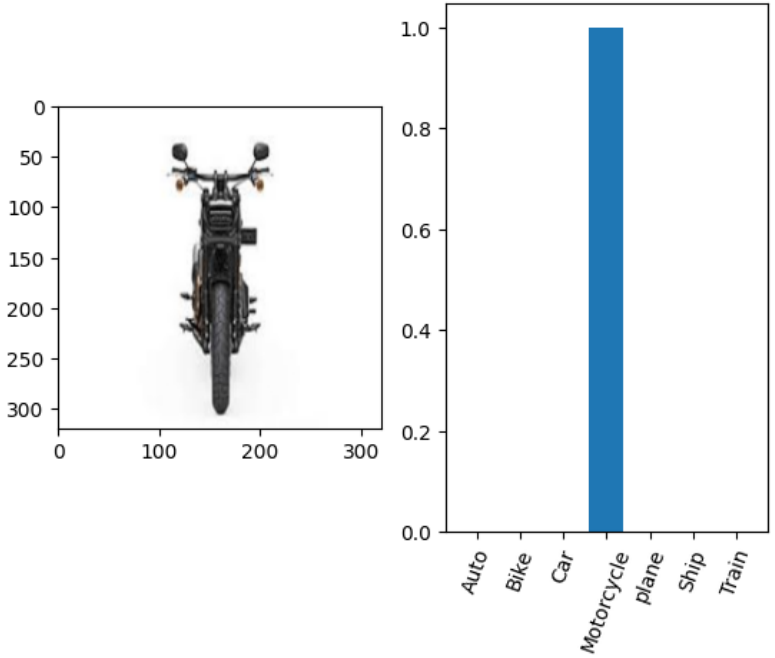
- Car
- Auto
- Bike
- Motorcycle
- Plane
- Ship
- Train

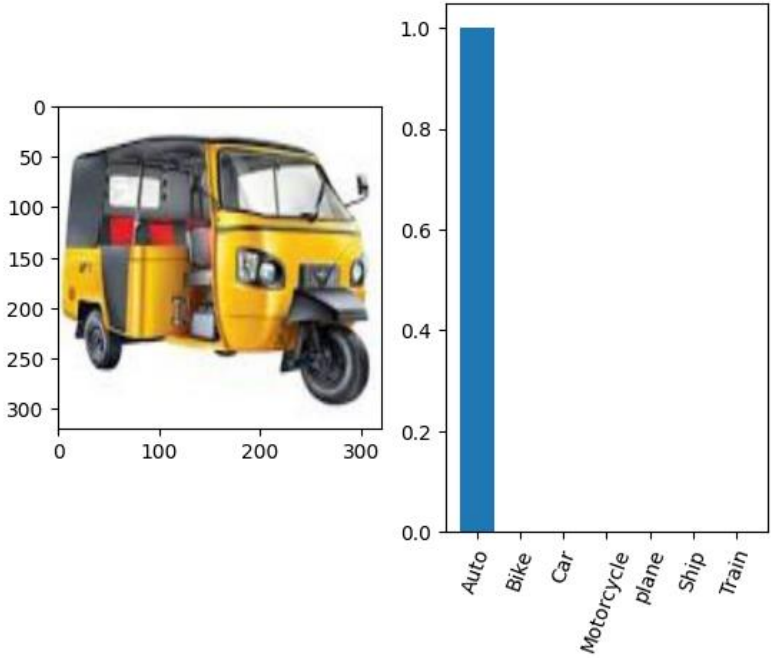
Outsource Images:

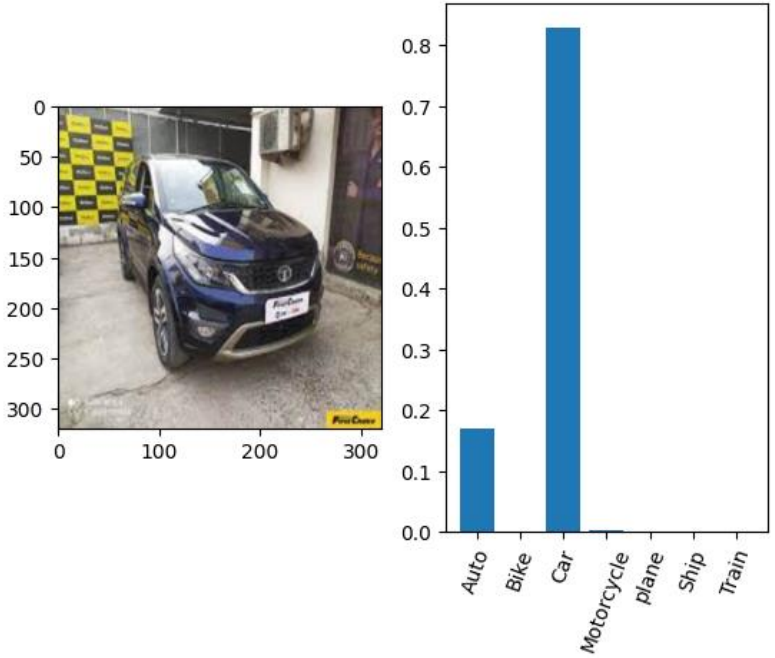
We have used image for personal items and images from internet:

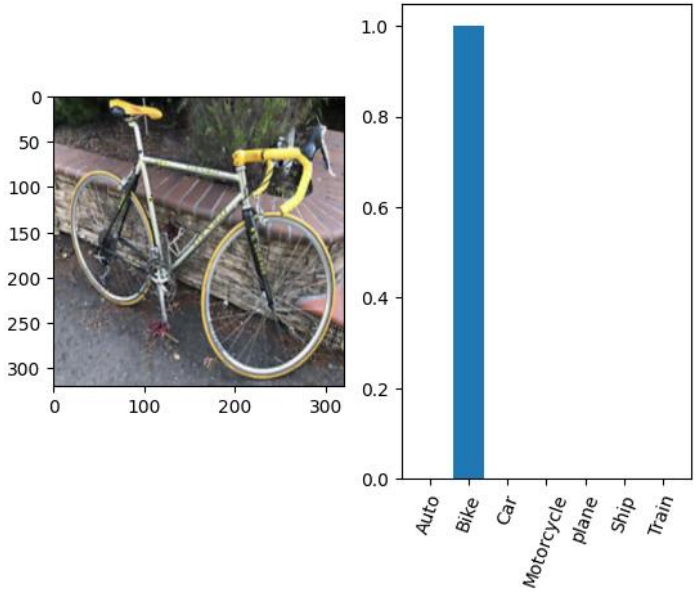
- Indian Auto
- Trains from Egypt
- Planes from Egypt
- Ship in Egypt
- Personal motorcycle

Internally from dataset:





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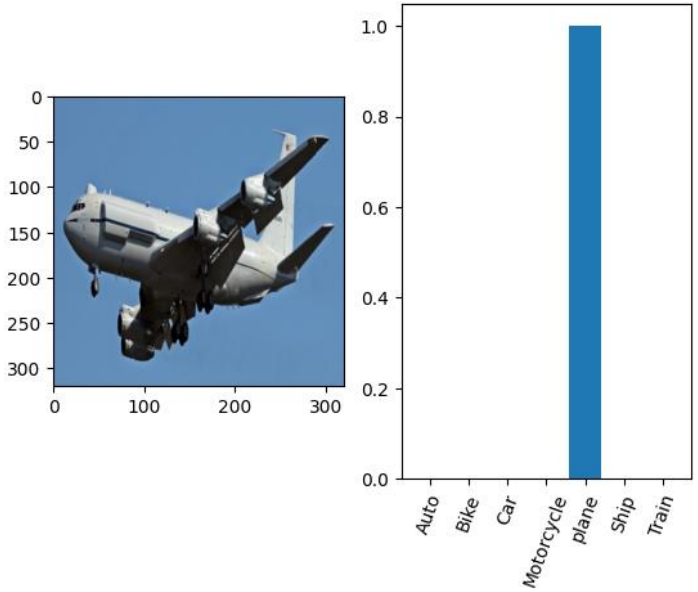
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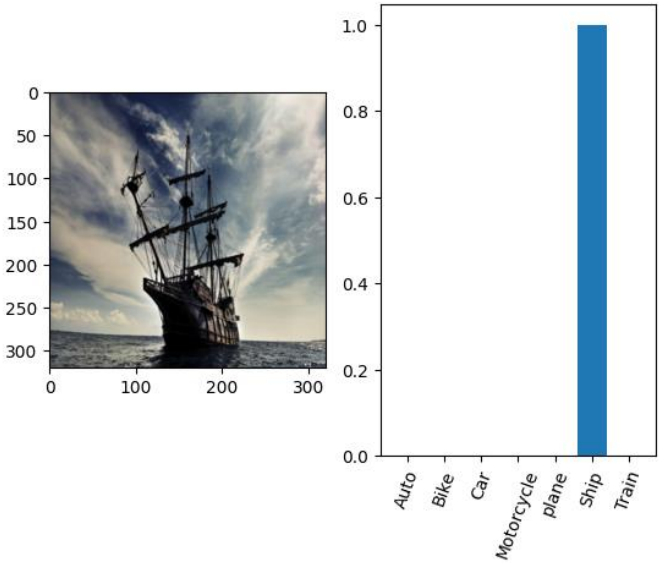
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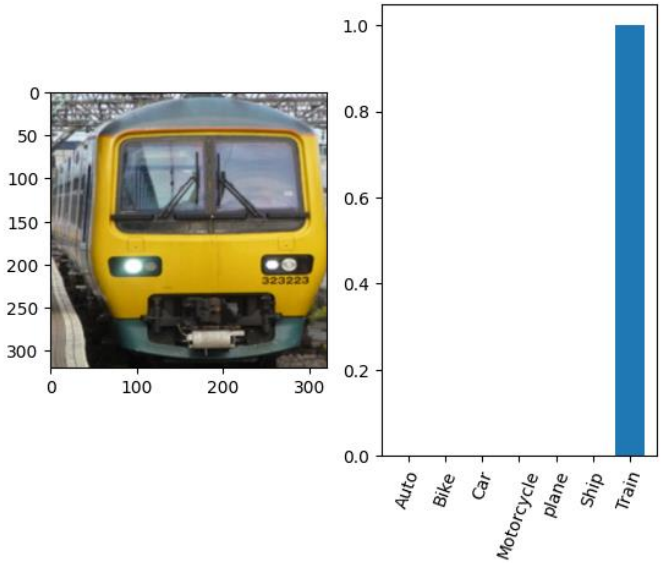
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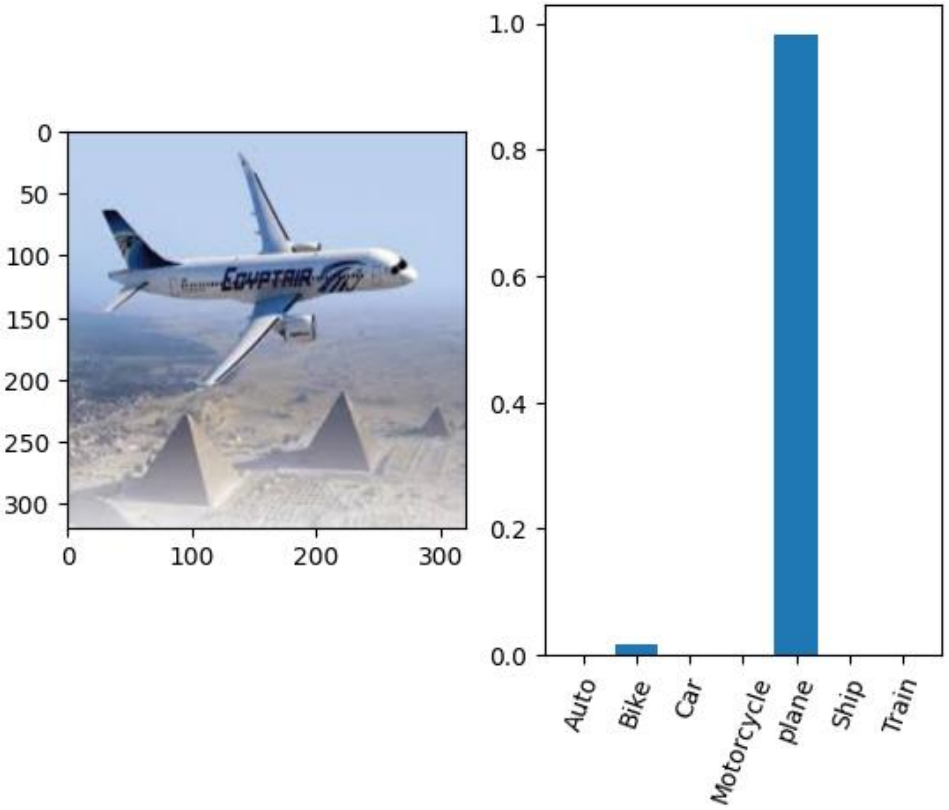

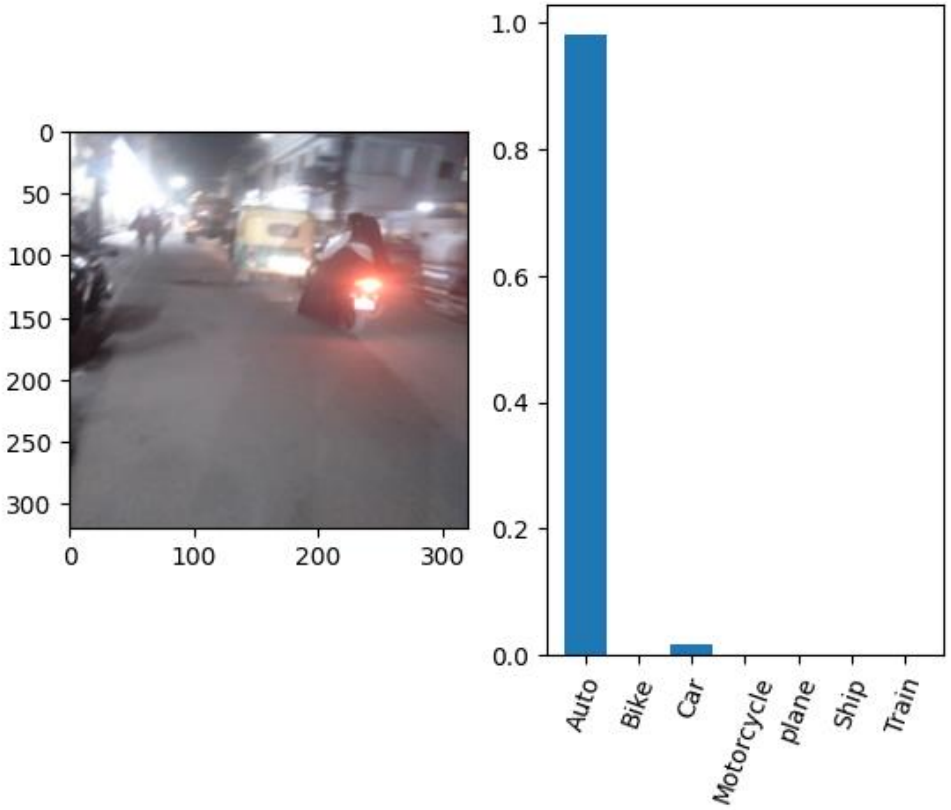

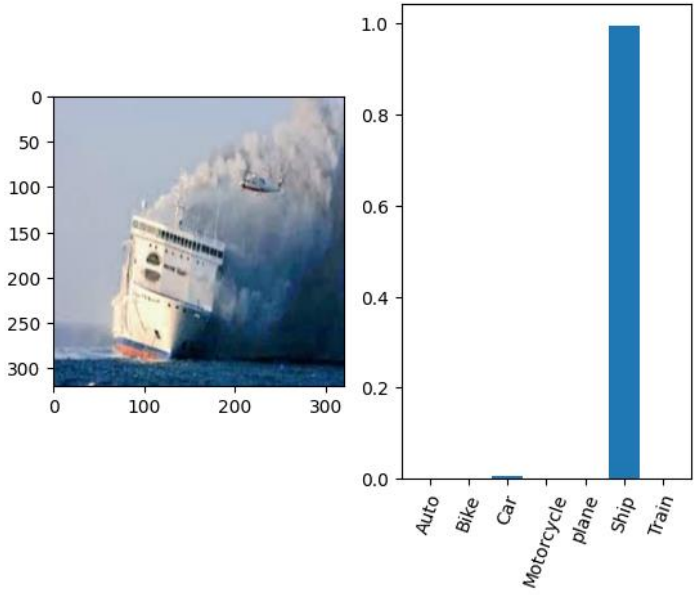
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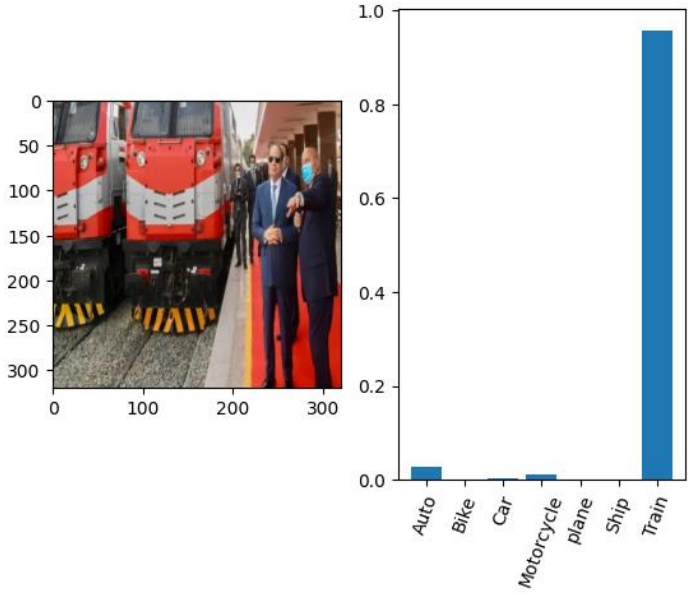
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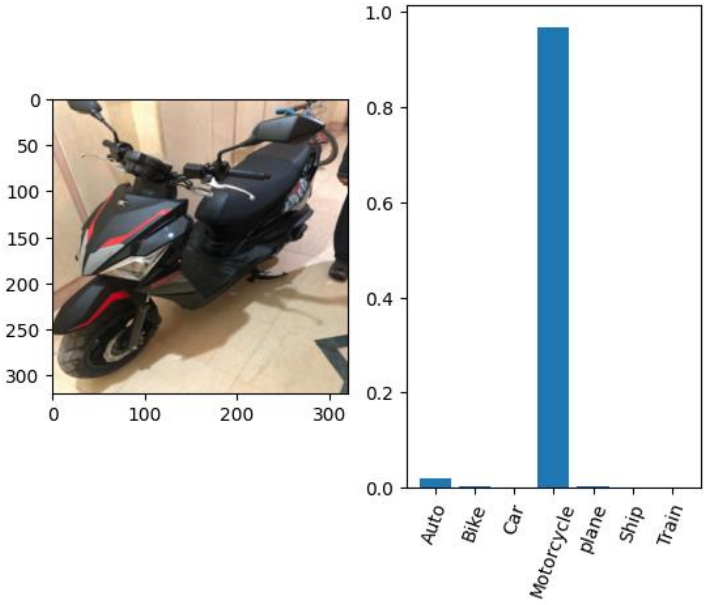
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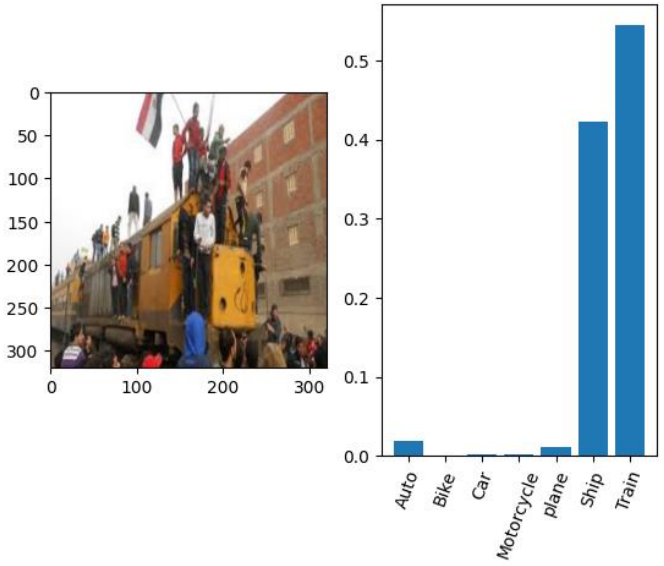
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Real Case Study

Transpo Tracker by StellarView.ai

https://www.youtube.com/watch?v=JNuYSOn-VnQ

Thanks for your listening!
Any Question?